United States Patent	12418267
Kind Code	B2
Date of Patent	September 16, 2025
Inventor(s)	Elgousi; Muhammed et al.

Harmonic trapping techniques for transmitter interstage matching

Abstract

A method for harmonic trapping in a matching network of a power amplifier includes determining primary inductance and secondary inductance of a differential transformer of the matching network, based on a signal operating frequency of the power amplifier. An inductance value for an L-C filter is determined based on the secondary inductance and a harmonic frequency of a local oscillator (LO) signal. A capacitance value for the L-C filter is determined based on the inductance value and the harmonic frequency of the LO signal. The L-C filter is provided on an electric connection between a direct current (DC) bias voltage source and a secondary inductor of the differential transformer. The L-C filter is configured with the determined inductance value and the determined capacitance value.

Inventors: **Elgousi; Muhammed (Chandler, AZ), Yu; Chuanzhao (Phoenix, AZ), Yoon; Hyun (Gilbert, AZ), Li; Xi (Chandler, AZ), Chang; LiChung Tony (Hillsboro, OR)**

Applicant: **Intel Corporation** (Santa Clara, CA)

Family ID: **1000008821050**

Assignee: **Intel Corporation (Santa Clara, CA)**

Appl. No.: **17/559341**

Filed: **December 22, 2021**

Prior Publication Data

Document Identifier	Publication Date
US 20230198479 A1	Jun. 22, 2023

Publication Classification

Int. Cl.: **H03F3/21** (20060101); **H03F1/26** (20060101); **H03F3/00** (20060101)

U.S. Cl.:

CPC **H03F3/21** (20130101); **H03F1/26** (20130101); **H03F3/005** (20130101); H03F2200/372 (20130101)

Field of Classification Search

CPC: H03F (3/21); H03F (1/26); H03F (3/005); H03F (2200/372); H03F (2200/451); H03F (3/193); H03F (1/565); H03F (3/245)

USPC: 330/250; 330/262-261; 330/302

References Cited

U.S. PATENT DOCUMENTS

Patent No.	Issued Date	Patentee Name	U.S. Cl.	CPC
8112044	12/2011	Marholev	455/90.3	H04B 5/48

FOREIGN PATENT DOCUMENTS

Patent No.	Application Date	Country	CPC
102022133755	12/2022	DE	N/A

Primary Examiner: Nguyen; Hieu P

Attorney, Agent or Firm: Schwegman Lundberg & Woessner, P.A.

Background/Summary

TECHNICAL FIELD
(1) Various embodiments generally may relate to the field of radio frequency (RF) signal communication by a wireless device, including harmonic trapping techniques for transmitter interstage matching.
BACKGROUND
(2) With the increased popularity of wireless communications such as fifth-generation (5G) communications and subsequent sixth-generation (6G) communications and beyond, there is a growing demand for increased flexibility and efficiency in using the communication bandwidth by wireless devices, including reducing non-linearities and interference signals associated with signal processing in transceiver systems of wireless devices.

Description

BRIEF DESCRIPTION OF THE DRAWINGS
(1) In the drawings, like numerals may describe the same or similar components or features in different views. Like numerals having different letter suffixes may represent different instances of similar components. Some embodiments are illustrated by way of example, and not limitation, in the figures of the accompanying drawings in which:
(2) FIG. **1** is a block diagram of a radio architecture with a harmonic trapping circuit, in accordance with some embodiments;

(3) FIG. **2** illustrates a front-end module circuitry for use in the radio architecture of FIG. **1**, in accordance with some embodiments;
(4) FIG. **3** illustrates a radio IC circuitry for use in the radio architecture of FIG. **1**, in accordance with some embodiments:
(5) FIG. **4** illustrates a baseband processing circuitry for use in the radio architecture of FIG. **1**, in accordance with some embodiments;
(6) FIG. **5** illustrates a block diagram of RF circuitry including mixer core circuitry driving a power amplifier (PA) circuitry, in accordance with some embodiments;
(7) FIG. **6** illustrates a block diagram of a more detailed view of the RF circuitry of FIG. **5**, in accordance with some embodiments;
(8) FIG. **7** illustrates a block diagram of PA input impedance at common mode, in accordance with some embodiments;
(9) FIG. **8** illustrates a block diagram of RF bias implementations in connection with the RF circuitry of FIG. **5**, in accordance with some embodiments;
(10) FIG. **9** illustrates a block diagram of RF bias implementations in connection with the RF circuitry of FIG. **5**, including multi-pole and adjustable pole harmonic filters, in accordance with some embodiments;
(11) FIG. **10** is a graph of common-mode trans-impedance showing the effect of a harmonic rejection filter, in accordance with some embodiments;
(12) FIG. **11** is a graph of feedback impedance showing the effect of a harmonic rejection filter, in accordance with some embodiments;
(13) FIG. **12** is a diagram illustrating common-mode feedback impedance at the PA of the RF circuitry of FIG. **6**, in accordance with some embodiments;
(14) FIG. **13** is a graph of simulated common-mode feedback impedance (Zd) associated with the RF circuitry in FIG. **12**, in accordance with some embodiments;
(15) FIG. **14** illustrates a flow diagram of a method for harmonic trapping in a matching network of a power amplifier, in accordance with some embodiments; and
(16) FIG. **15** illustrates a block diagram of an example machine upon which any one or more of the operations/techniques (e.g., methodologies) discussed herein may perform.

DETAILED DESCRIPTION

(17) The following detailed description refers to the accompanying drawings. The same reference numbers may be used in different drawings to identify the same or similar elements. In the following description, for purposes of explanation and not limitation, specific details are set forth such as particular structures, architectures, interfaces, techniques, etc. to provide a thorough understanding of the various aspects of various embodiments. However, it will be apparent to those skilled in the art having the benefit of the present disclosure that the various aspects of the various embodiments may be practiced in other examples that depart from these specific details. In certain instances, descriptions of well-known devices, circuits, and methods are omitted so as not to obscure the description of the various embodiments with unnecessary detail.
(18) The following description and the drawings sufficiently illustrate specific embodiments to enable those skilled in the art to practice them. Other embodiments may incorporate structural, logical, electrical, process, and other changes. Portions and features of some embodiments may be included in or substituted for, those of other embodiments. Embodiments outlined in the claims encompass all available equivalents of those claims.
(19) In some embodiments, due to the differential nature of in-phase (I) and quadrature (Q) mixing with a differential local oscillator (LO) frequency, even-order harmonics of the LO signal will be generated. In low-frequency LO systems (e.g., wireless systems operating below 10 GHz), the 4th order harmonic noise power generates a spur at the main channel for millimeter-wave (mmWave) applications, such as 5G and beyond wireless systems.
(20) The disclosed techniques include using a harmonic trapping circuit such as a filter

arrangement (e.g., at least one L-C filter) located between the matching circuitry transformer and the output of a direct current (DC) bias voltage source of the power amplifier (PA). Usage of the disclosed techniques (e.g., the disclosed filter arrangement which can include an LO noise rejection filter such as an L-C filter for one or more LO signal harmonics) can be ascertained by visual inspection of the matching network (e.g., via micrograph of the die at the input of the PA matching circuitry). The LO noise rejection filter inductor can be implemented within the periphery of the PA input transformer to save area and reduce routing parasitics, and usage of the disclosed techniques may be assessed by PA input transformer inspection.

(21) Additional functionalities of the harmonic trapping circuit are discussed in connection with at least FIGS. **1**, **2**, and **5-14**.

(22) FIG. **1** is a block diagram of a radio architecture **100** with a harmonic trapping circuit, in accordance with some embodiments. The radio architecture **100** may be implemented in a computing device (e.g., device **1500** in FIG. **15**) including user equipment (UE), a base station (e.g., a next generation Node-B (gNB), enhanced Node-B (eNB)), a smartphone, or another type of wired or wireless device. The radio architecture **100** may include radio front-end module (FEM) circuitry **104**, radio integrated circuit (IC) circuitry **106**, and baseband processing circuitry **108**. Radio architecture **100** as shown includes both Wireless Local Area Network (WLAN) functionality and Bluetooth (BT) functionality although embodiments are not so limited and the disclosed techniques apply to other types of radio architectures as well. In this disclosure, “WLAN” and “Wi-Fi” are used interchangeably. In some aspects, a single wireless functionality can be configured (e.g., one of WLAN, BT, or another type of wires functionality).

(23) FEM circuitry **104** may include a WLAN or Wi-Fi FEM circuitry **104**A and a Bluetooth (BT) FEM circuitry **104**B. The WLAN FEM circuitry **104**A may include a receive signal path comprising circuitry configured to operate on WLAN RF signals received from one or more antennas **101**, to amplify the received signals, and to provide the amplified versions of the received signals to the WLAN radio IC circuitry **106**A for further processing. The BT FEM circuitry **104**B may include a receive signal path which may include circuitry configured to operate on BT RF signals received from the one or more antennas **101**, to amplify the received signals and to provide the amplified versions of the received signals to the BT radio IC circuitry **106**B for further processing. The FEM circuitry **104**A may also include a transmit signal path which may include circuitry configured to amplify WLAN signals provided by the radio IC circuitry **106**A for wireless transmission by the one or more antennas **101**. Besides, the FEM circuitry **104**B may also include a transmit signal path which may include circuitry configured to amplify BT signals provided by the radio IC circuitry **106**B for wireless transmission by the one or more antennas. In the embodiment of FIG. **1**, although FEM circuitry **104**A and FEM circuitry **104**B are shown as being distinct from one another, embodiments are not so limited and include within their scope the use of a FEM (not shown) that includes a transmit path and/or a receive path for both WLAN and BT signals, or the use of one or more FEM circuitries where at least some of the FEM circuitries share transmit and/or receive signal paths for both WLAN and BT signals.

(24) Radio IC circuitry **106** as shown may include WLAN radio IC circuitry **106**A and BT radio IC circuitry **106**B. The WLAN radio IC circuitry **106**A may include a receive signal path which may include circuitry to down-convert WLAN RF signals received from the FEM circuitry **104**A and provide baseband signals to WLAN baseband processing circuitry **108**A. The BT radio IC circuitry **106**B may, in turn, include a receive signal path which may include circuitry to down-convert BT RF signals received from the FEM circuitry **104**B and provide baseband signals to BT baseband processing circuitry **108**B. The WLAN radio IC circuitry **106**A may also include a transmit signal path which may include circuitry to up-convert WLAN baseband signals provided by the WLAN baseband processing circuitry **108**A and provide WLAN RF output signals to the FEM circuitry **104**A for subsequent wireless transmission by the one or more antennas **101**. The BT radio IC circuitry **106**B may also include a transmit signal path which may include circuitry to up-convert

BT baseband signals provided by the BT baseband processing circuitry **108**B and provide BT RF output signals to the FEM circuitry **104**B for subsequent wireless transmission by the one or more antennas **101**. In the embodiment of FIG. **1**, although radio IC circuitries **106**A and **106**B are shown as being distinct from one another, embodiments are not so limited and include within their scope the use of a radio IC circuitry (not shown) that includes a transmit signal path and/or a receive signal path for both WLAN and BT signals, or the use of one or more radio IC circuitries where at least some of the radio IC circuitries share transmit and/or receive signal paths for both WLAN and BT signals.

(25) Baseband processing circuitry **108** may include a WLAN baseband processing circuitry **108**A and a BT baseband processing circuitry **108**B. The WLAN baseband processing circuitry **108**A may include a memory, such as, for example, a set of RAM arrays in a Fast Fourier Transform (FFT) or Inverse Fast Fourier Transform (IFFT) block (not shown) of the WLAN baseband processing circuitry **108**A. Each of the WLAN baseband processing circuitry **108**A and the BT baseband processing circuitry **108**B may further include one or more processors and control logic to process the signals received from the corresponding WLAN or BT receive signal path of the radio IC circuitry **106**, and to also generate corresponding WLAN or BT baseband signals for the transmit signal path of the radio IC circuitry **106**. Each of the baseband processing circuitries **108**A and **108**B may further include physical layer (PHY) and medium access control layer (MAC) circuitry and may further interface with the application processor **111** for generation and processing of the baseband signals and for controlling operations of the radio IC circuitry **106**.

(26) In some embodiments, radio architecture **100** includes a harmonic trapping (HT) circuit **105** configured to filter out one or more harmonics (e.g., even-order harmonics) of a local oscillator (LO) signal. In some embodiments, the HT circuit **105** can be configured as part of the FEM circuitry **104** (e.g., before the PA as illustrated in FIG. **2**) or as part of the radio IC circuitry **106** (e.g., after the mixer as illustrated in FIG. **3**). The functionality of the HT circuit is discussed in greater detail in connection with FIGS. **5-14**.

(27) Referring still to FIG. **1**, according to the shown embodiment, WLAN-BT coexistence circuitry **113** may include logic providing an interface between the WLAN baseband processing circuitry **108**A and the BT baseband processing circuitry **108**B to enable use cases requiring WLAN and BT coexistence. In addition, a switch **103** may be provided between the WLAN FEM circuitry **104**A and the BT FEM circuitry **104**B to allow switching between the WLAN and BT radios according to application needs. In addition, although the one or more antennas **101** are depicted as being respectively connected to the WLAN FEM circuitry **104**A and the BT FEM circuitry **104**B, embodiments include within their scope the sharing of the one or more antennas **101** as between the WLAN and BT FEMs, or the provision of more than one antenna connected to each of FEM **104**A or **104**B.

(28) In some embodiments, the front-end module circuitry **104**, the radio IC circuitry **106**, and the baseband processing circuitry **108** may be provided on a single radio card, such as wireless radio card **102**. In some other embodiments, the one or more antennas **101**, the FEM circuitry **104**, and the radio IC circuitry **106** may be provided on a single radio card. In some other embodiments, the radio IC circuitry **106** and the baseband processing circuitry **108** may be provided on a single chip or IC, such as IC **112**.

(29) In some embodiments, the wireless radio card **102** may include a WLAN radio card and may be configured for Wi-Fi communications, although the scope of the embodiments is not limited in this respect. In some of these embodiments, the radio architecture **100** may be configured to receive and transmit orthogonal frequency division multiplexed (OFDM) or orthogonal frequency division multiple access (OFDMA) communication signals over a multicarrier communication channel. The OFDM or OFDMA signals may comprise a plurality of orthogonal subcarriers. In some embodiments, the wireless radio card **102** may include a platform controller hub (PCH) system-on-a-chip (SOC) and a central processing unit (CPU)/host SOC.

(30) In some of these multicarrier embodiments, radio architecture **100** may be part of a Wi-Fi communication station (STA) such as a wireless access point (AP), a base station, or a mobile device including a Wi-Fi enabled device. In some of these embodiments, radio architecture **100** may be configured to transmit and receive signals in accordance with specific communication standards and/or protocols, such as any of the Institute of Electrical and Electronics Engineers (IEEE) standards including, 802.11n-2009, IEEE 802.11-2012, 802.11n-2009, 802.11ac, IEEE 802.11-2016, and/or 802.11ax standards and/or proposed specifications for WLANs, although the scope of embodiments is not limited in this respect. Radio architecture **100** may also be suitable to transmit and/or receive communications in accordance with other techniques and standards, including a 3.sup.rd Generation Partnership Project (3GPP) standard, including a communication standard used in connection with 5G or new radio (NR) communications.
(31) In some embodiments, the radio architecture **100** may be configured for high-efficiency (HE) Wi-Fi communications in accordance with the IEEE 802.1 lax standard or another standard associated with wireless communications. In these embodiments, the radio architecture **100** may be configured to communicate in accordance with an OFDMA technique, although the scope of the embodiments is not limited in this respect.
(32) In some other embodiments, the radio architecture **100** may be configured to transmit and receive signals transmitted using one or more other modulation techniques such as spread spectrum modulation (e.g., direct sequence code division multiple access (DS-CDMA) and/or frequency hopping code division multiple access (FH-CDMA)), time-division multiplexing (TDM) modulation, and/or frequency-division multiplexing (FDM) modulation, although the scope of the embodiments is not limited in this respect.
(33) In some embodiments, as further shown in FIG. **1**, the BT baseband processing circuitry **108**B may be compliant with a Bluetooth (BT) connectivity standard such as Bluetooth, Bluetooth 4.0 or Bluetooth 5.0, or any other iteration of the Bluetooth Standard. In embodiments that include BT functionality as shown for example in FIG. **1**, the radio architecture **100** may be configured to establish a BT synchronous connection-oriented (SCO) link and or a BT low energy (BT LE) link. In some of the embodiments that include functionality, the radio architecture **100** may be configured to establish an extended SCO (eSCO) link for BT communications, although the scope of the embodiments is not limited in this respect. In some of these embodiments that include a BT functionality, the radio architecture may be configured to engage in a BT Asynchronous Connection-Less (ACL) communications, although the scope of the embodiments is not limited in this respect. In some embodiments, as shown in FIG. **1**, the functions of a BT radio card and WLAN radio card may be combined on a single wireless radio card, such as the single wireless radio card **102**, although embodiments are not so limited, and include within their scope discrete WLAN and BT radio cards
(34) In some embodiments, the radio architecture **100** may include other radio cards, such as a cellular radio card configured for cellular (e.g., 3GPP such as LTE, LTE-Advanced, or 5G communications).
(35) In some IEEE 802.11 embodiments, the radio architecture **100** may be configured for communication over various channel bandwidths including bandwidths having center frequencies of about 900 MHz, 2.4 GHz, 5 GHz, and bandwidths of about 1 MHz, 2 MHz, 2.5 MHz, 4 MHz, 5 MHz, 8 MHz, 10 MHz, 16 MHz, 20 MHz, 40 MHz, 80 MHz (with contiguous bandwidths) or 80+80 MHz (160 MHz) (with non-contiguous bandwidths). In some embodiments, a 320 MHz channel bandwidth may be used. The scope of the embodiments is not limited with respect to the above center frequencies, however.
(36) FIG. **2** illustrates FEM circuitry **200** in accordance with some embodiments. The FEM circuitry **200** is one example of circuitry that may be suitable for use as the WLAN and/or BT FEM circuitry **104**A/**104**B (FIG. **1**), although other circuitry configurations may also be suitable.
(37) In some embodiments, the FEM circuitry **200** may include a TX/RX switch **202** to switch

between transmit (TX) mode and receive (RX) mode operation. In some aspects, a diplexer may be used in place of a TX/RX switch. The FEM circuitry **200** may include a receive signal path and a transmit signal path. The receive signal path of the FEM circuitry **200** may include a low-noise amplifier (LNA) **206** to amplify received RF signals **203** and provide the amplified received RF signals **207** as an output (e.g., to the radio IC circuitry **106** (FIG. **1**)). The transmit signal path of the FEM circuitry **200** may include a power amplifier (PA) **210** to amplify input RF signals **209** (e.g., provided by the radio IC circuitry **106**), and one or more filters **212**, such as band-pass filters (BPFs), low-pass filters (LPFs) or other types of filters, to generate RF signals **215** for subsequent transmission (e.g., by the one or more antennas **101** in FIG. **1**).

(38) In some dual-mode embodiments for Wi-Fi communication, the FEM circuitry **200** may be configured to operate in, e.g., either the 2.4 GHz frequency spectrum or the 5 GHz frequency spectrum. In these embodiments, the receive signal path of the FEM circuitry **200** may include a receive signal path duplexer **204** to separate the signals from each spectrum as well as provide a separate LNA **206** for each spectrum as shown. In these embodiments, the transmit signal path of the FEM circuitry **200** may also include a PA **210** and one or more filters **212**, such as a BPF, an LPF, or another type of filter for each frequency spectrum, and a transmit signal path duplexer **214** to provide the signals of one of the different spectrums onto a single transmit path for subsequent transmission by the one or more antennas **101** (FIG. **1**). In some embodiments, BT communications may utilize the 2.4 GHz signal paths and may utilize the same FEM circuitry **200** as the one used for WLAN communications.

(39) In some embodiments, the input RF signals **209** can be configured as an input to HT circuit **105**. In some embodiments, the input RF signal **209** is a differential baseband signal generated based on an LO signal (e.g., LO signal generated by synthesizer circuitry **304** in FIG. **3**). The mixer (e.g., mixer circuitry **314**) can drive the PA **210** using the differential baseband signal which is received by a transformer of the PA **210**. In some embodiments, the HT circuit **105** can be a filter arrangement within the FEM circuitry **104**, placed between a DC bias voltage source of the PA **210** and the transformer, to filter even-order harmonics associated with the LO signal and present within the differential baseband signal.

(40) FIG. **3** illustrates radio IC circuitry **300** in accordance with some embodiments. The radio IC circuitry **300** is one example of circuitry that may be suitable for use as the WLAN or BT radio IC circuitry **106**A/**106**B (FIG. **1**), although other circuitry configurations may also be suitable.

(41) In some embodiments, the radio IC circuitry **300** may include a receive signal path and a transmit signal path. The receive signal path of the radio IC circuitry **300** may include mixer circuitry **302**, such as, for example, down-conversion mixer circuitry, amplifier circuitry **306**, and filter circuitry **308**. The transmit signal path of the radio IC circuitry **300** may include at least filter circuitry **312** and mixer circuitry **314**, such as up-conversion mixer circuitry. Radio IC circuitry **300** may also include synthesizer circuitry **304** for synthesizing a frequency **305** for use by the mixer circuitry **302** and the mixer circuitry **314**. The mixer circuitry **302** and/or **314** may each, according to some embodiments, be configured to provide direct conversion functionality. The latter type of circuitry presents a much simpler architecture as compared with standard super-heterodyne mixer circuitries, and any flicker noise brought about by the same may be alleviated for example through the use of OFDM modulation. FIG. **3** illustrates only a simplified version of a radio IC circuitry and may include, although not shown, embodiments where each of the depicted circuitries may include more than one component. For instance, mixer circuitry **302** and/or **314** may each include one or more mixers, and filter circuitries **308** and/or **312** may each include one or more filters, such as one or more BPFs and/or LPFs according to application needs. For example, when mixer circuitries are of the direct-conversion type, they may each include two or more mixers.

(42) In some embodiments, mixer circuitry **302** may be configured to down-convert RF signals **207** received from the FEM circuitry **104** (FIG. **1**) based on the synthesized frequency **305** provided by the synthesizer circuitry **304**. The amplifier circuitry **306** may be configured to amplify the down-

converted signals and the filter circuitry **308** may include an LPF configured to remove unwanted signals from the down-converted signals to generate output baseband signals **307**. Output baseband signals **307** may be provided to the baseband processing circuitry **108** (FIG. **1**) for further processing. In some embodiments, the output baseband signals **307** may be zero-frequency baseband signals, although this is not a requirement. In some embodiments, mixer circuitry **302** may comprise passive mixers, although the scope of the embodiments is not limited in this respect.
(43) In some embodiments, the mixer circuitry **314** may be configured to up-convert input baseband signals **311** based on the synthesized frequency **305** provided by the synthesizer circuitry **304** to generate input RF signals **209** for the FEM circuitry **104**. The baseband signals **311** may be provided by the baseband processing circuitry **108** and may be filtered by filter circuitry **312**. The filter circuitry **312** may include an LPF or a BPF, although the scope of the embodiments is not limited in this respect.
(44) In some embodiments, the mixer circuitry **302** and the mixer circuitry **314** may each include two or more mixers and may be arranged for quadrature down-conversion and/or up-conversion respectively with the help of the synthesizer circuitry **304**. In some embodiments, the mixer circuitry **302** and the mixer circuitry **314** may each include two or more mixers each configured for image rejection (e.g., Hartley image rejection). In some embodiments, the mixer circuitry **302** and the mixer circuitry **314** may be arranged for direct down-conversion and/or direct up-conversion, respectively. In some embodiments, the mixer circuitry **302** and the mixer circuitry **314** may be configured for super-heterodyne operation, although this is not a requirement.
(45) In some embodiments, the HT circuit **105** can be configured as part of the radio IC circuitry **300** and can be placed after the mixer circuitry **314**, at the input of the PA **210**. Additional functionalities of the HT circuit are discussed in connection with FIGS.
(46) Mixer circuitry **302** may comprise, according to one embodiment: quadrature passive mixers (e.g., for the in-phase (I) and quadrature-phase (Q) paths). In such an embodiment, RF input signal **207** from FIG. **2** may be down-converted to provide I and Q baseband output signals to be sent to the baseband processor.
(47) Quadrature passive mixers may be driven by zero and ninety-degree time-varying LO switching signals provided by a quadrature circuitry which may be configured to receive a LO frequency (fLO) from a local oscillator or a synthesizer, such as synthesized frequency **305** of synthesizer circuitry **304** (FIG. **3**). In some embodiments, the LO frequency may be the carrier frequency, while in other embodiments, the LO frequency may be a fraction of the carrier frequency (e.g., one-half the carrier frequency, one-third the carrier frequency). In some embodiments, the zero and ninety-degree time-varying switching signals may be generated by the synthesizer, although the scope of the embodiments is not limited in this respect.
(48) In some embodiments, the LO signals may differ in the duty cycle (the percentage of one period in which the LO signal is high) and/or offset (the difference between start points of the period). In some embodiments, the LO signals may have a 25% duty cycle and a 50% offset. In some embodiments, each branch of the mixer circuitry (e.g., the in-phase (I) and quadrature-phase (Q) path) may operate at a 25% duty cycle, which may result in a significant reduction in power consumption.
(49) The RF input signal **207** (FIG. **2**) may comprise a balanced signal, although the scope of the embodiments is not limited in this respect. The I and Q baseband output signals may be provided to the low-noise amplifier, such as amplifier circuitry **306** (FIG. **3**) or to filter circuitry **308** (FIG. **3**).
(50) In some embodiments, the output baseband signals **307** and the input baseband signals **311** may be analog, although the scope of the embodiments is not limited in this respect. In some alternate embodiments, the output baseband signals **307** and the input baseband signals **311** may be digital. In these alternate embodiments, the radio IC circuitry may include an analog-to-digital converter (ADC) and digital-to-analog converter (DAC) circuitry.
(51) In some dual-mode embodiments, a separate radio IC circuitry may be provided for processing

signals for each spectrum, or for other spectrums not mentioned here, although the scope of the embodiments is not limited in this respect.

(52) In some embodiments, the synthesizer circuitry **304** may be a fractional-N synthesizer or a fractional N/N+1 synthesizer, although the scope of the embodiments is not limited in this respect as other types of frequency synthesizers may be suitable. In some embodiments, the synthesizer circuitry **304** may be a delta-sigma synthesizer, a frequency multiplier, or a synthesizer comprising a phase-locked loop with a frequency divider. According to some embodiments, the synthesizer circuitry **304** may include a digital frequency synthesizer circuitry. An advantage of using a digital synthesizer circuitry is that, although it may still include some analog components, its footprint may be scaled down much more than the footprint of an analog synthesizer circuitry. In some embodiments, frequency input into synthesizer circuitry **304** may be provided by a voltage-controlled oscillator (VCO), although that is not a requirement. A divider control input may further be provided by either the baseband processing circuitry **108** (FIG. **1**) or the application processor **111** (FIG. **1**) depending on the desired output frequency **305**. In some embodiments, a divider control input (e.g., N) may be determined from a look-up table (e.g., within a Wi-Fi card) based on a channel number and a channel center frequency as determined or indicated by the application processor **11**.

(53) In some embodiments, synthesizer circuitry **304** may be configured to generate a carrier frequency as the output frequency **305**, while in other embodiments, the output frequency **305** may be a fraction of the carrier frequency (e.g., one-half the carrier frequency, one-third the carrier frequency). In some embodiments, the output frequency **305** may be a LO frequency (fLO).

(54) FIG. **4** illustrates a functional block diagram of baseband processing circuitry **400** in accordance with some embodiments. The baseband processing circuitry **400** is one example of circuitry that may be suitable for use as the baseband processing circuitry **108** (FIG. **1**), although other circuitry configurations may also be suitable. The baseband processing circuitry **400** may include a receive baseband processor (RX BBP) **402** for processing received analog baseband signals **309** provided by the radio IC circuitry **106** (FIG. **1**) and a transmit baseband processor (TX BBP) **404** for generating analog input baseband signals **311** for the radio IC circuitry **106**. The baseband processing circuitry **400** may also include control logic **406** for coordinating the operations of the baseband processing circuitry **400**.

(55) In some embodiments (e.g., when analog baseband signals are exchanged between the baseband processing circuitry **400** and the radio IC circuitry **106**), the baseband processing circuitry **400** may include an analog-to-digital converter (ADC) **410** to convert analog baseband signals **309** received from the radio IC circuitry **106** to digital baseband signals for processing by the RX BBP **402**. In these embodiments, the baseband processing circuitry **400** may also include a digital-to-analog converter (DAC) **408** to convert digital baseband signals from the TX BBP **404** to analog input baseband signals **311**.

(56) In some embodiments that communicate OFDM signals or OFDMA signals, such as through the WBPC **108**A, the TX BBP **404** may be configured to generate OFDM or OFDMA signals as appropriate for transmission by performing an inverse fast Fourier transform (IFFT). The RX BBP **402** may be configured to process received OFDM signals or OFDMA signals by performing an FFT. In some embodiments, the RX BBP **402** may be configured to detect the presence of an OFDM signal or OFDMA signal by performing an autocorrelation, to detect a preamble, such as a short preamble, and by performing a cross-correlation, to detect a long preamble. The preambles may be part of a predetermined frame structure for Wi-Fi communication.

(57) Referring back to FIG. **1**, in some embodiments, the one or more antennas **101** (FIG. **1**) may each comprise one or more directional or omnidirectional antennas, including, for example, dipole antennas, monopole antennas, patch antennas, loop antennas, microstrip antennas or other types of antennas suitable for transmission of RF signals. In some multiple-input multiple-output (MIMO) embodiments, the antennas may be effectively separated to take advantage of spatial diversity and

the different channel characteristics that may result. The one or more antennas **101** may each include a set of phased-array antennas, although embodiments are not so limited.

(58) Although the radio architecture **100** is illustrated as having several separate functional elements, one or more of the functional elements may be combined and may be implemented by combinations of software configured elements, such as processing elements including digital signal processors (DSPs), and/or other hardware elements. For example, some elements may comprise one or more microprocessors, DSPs, field-programmable gate arrays (FPGAs), application-specific integrated circuits (ASICs), radio-frequency integrated circuits (RFICs), and combinations of various hardware and logic circuitry for performing at least the functions described herein. In some embodiments, the functional elements may refer to one or more processes operating on one or more processing elements.

(59) FIG. **5** illustrates a block diagram of RF circuitry **500** including mixer core circuitry driving a power amplifier (PA) circuitry, in accordance with some embodiments. Referring to FIG. **5**, RF circuitry **500** includes a mixer core **502**, matching circuitry (or matching network) **504**, PA **506**, and output balun **508**.

(60) The mixer core **502** includes DACs **510** and **512**, a local oscillator (or synthesizer circuitry) **514**, mixers **516** and **518**, an adder **520**. The PA **506** can include a neutralized differential pair of transistors **522**. The matching circuitry **504** can be configured as a transformer coupling the output of the mixer core **502** (e.g., a differential analog signal generated by mixers **516** and **518**) to the input of PA **506**. In this regard, the transformer of the matching circuitry **504** can be used for driving the PA **506** by the differential signal output of mixer core **502**. The output of PA **506** is output to at least one antenna via the single-ended balun **508**.

(61) FIG. **6** illustrates a block diagram of a more detailed view of the RF circuitry of FIG. **5**, in accordance with some embodiments. Referring to FIG. **6**, RF circuitry **600** includes mixer core **602**, matching circuitry **604**, and PA **606**. Mixer core **602** can include a digital signal processor **608** (e.g., a baseband processor such as TX baseband processor **404** in baseband processing circuitry **400** of FIG. **4**), DACs **610** (e.g., DAC **408**), low-pass filters (LPFs) **612** (e.g., filter circuitry **312** of FIG. **3**), and mixers **614** (e.g., mixer circuitry **314** in FIG. **3**). Mixers **614** are configured to upconvert an RF signal using an LO signal (e.g., LO signal generated by synthesizer circuitry **304**). Matching circuitry **604** includes transformer **618** and HT circuit **628** (which can be the same as HT circuit **105** discussed previously). Transformer **618** can be a differential input transformer used for driving PA **606** via the differential baseband signal **616** generated as output by the mixer core **602**. PA **606** includes differential pair of transistors NN and NP generating signal outputs OutN and OutP. PA **606** further includes neutralization capacitors Cneut couple to the gates of transistors NN and NP.

(62) In an example embodiment, transformer **618** includes a primary inductor **620** coupled to the mixer core **602** and a secondary inductor **622** coupled to the input of PA **606**. A mixer DC supply **624** can be coupled to the primary inductor **620**, and a DC bias voltage source **626** can be coupled to the secondary inductor **622**.

(63) In an example embodiment, HT circuit **628** is coupled between the DC bias voltage source **626** and the secondary inductor **622** and is configured to filter out one or more harmonics of the LO signal used to generate the differential baseband signal **616**. In some aspects, the HT circuit **628** is configured as a filter arrangement including at least one L-C filter (e.g., inductor L.sub.HF and capacitor C.sub.HF) for filtering out one or more harmonics of the LO signal, such as an even-order harmonic (e.g., the fourth harmonic of the LO signal). In an example embodiment, HT circuit **628** further includes a grounded capacitor CL **630** configured to filter low-frequency noise signals generated by the DC bias voltage source **626**.

(64) In some embodiments associated with an RF direct conversion topology transceiver (e.g., including circuit topologies illustrated in FIG. **5** in FIG. **6**), a 90° phase-shifted I and Q differential baseband signal (e.g., differential baseband signal **616**) is generated and upconverted using an RF

signal generated using an LO signal. To improve the overall noise rejection of the RF circuitry **600**, the RF line-up can be designed in differential mode up to the final antenna interface, where a differential-to-single-ended balun (e.g., balun **508**) is used. Due to the differential nature of IQ mixing with differential LO frequency, even-order harmonics of the LO will be generated. In low-frequency wireless systems (usually below 10 GHz), the 4th order harmonic noise power ends up generating a spur at the main channel for mmWave applications (e.g., 5G and beyond). The source of this elevated LO 4th order harmonic spur is coming from the common-mode (CM) feedback noise coupling on the DC bias for the PA **606** that is driven by the mixer core **602**, as shown in FIG. **6**. In some aspects, this issue may be addressed by inserting a high decoupling capacitor at the center tap node of the secondary inductor **622** to short this high-frequency noise to the ground. However, this high-value capacitor, along with the addition of the differential pair parasitic capacitance, creates a low-frequency pole when resonating with the transformer inductors, causing a higher potential for PA instability and oscillation at common-mode.

(65) The common-mode noise generated by the feedback ends up modulating the bias node since the impedance looking from the PA input into the DC-bias node is very high, hence the noise is directly fed at the PA input and gets amplified by the PA **606**. Although the gain of the PA **606** is low at the 4th harmonic frequency region, the power level of the spur is high.

(66) In some embodiments, to reduce the power of the LO 4th order harmonic spur, HT circuit **628** can be used. More specifically, HT circuit **628** can be configured as an LC filter at the center tap of the secondary inductor **622** of transformer **618**, as shown in FIG. **6**. In some embodiments, the added LC filter can be designed to meet resonate at the 4th order harmonic of the LO (ω04). However, adding this LC filter could introduce a potential instability to PA **606** due to the feedback nature of the PA which is caused by the gate-drain capacitance (Cgd) and the neutralization capacitor (Cneut) in common mode. The design flow is explained hereinbelow.

(67) Since the inductor L.sub.HF of the L-C filter in HT circuit **628** is designed to address the 4th harmonic frequency of the LO, its value will be smaller compared to the interstage transformer inductor. Hence, such an inductor can be implemented within the interstage transformer matching, as shown in FIGS. **6**, **8**, and **9**. The quality factor of the filter inductor can be designed to be low to ensure no peaking is introduced at a lower frequency, where this inductor can resonate with the total decoupling capacitance of the RF bias in addition to the increased parasitic PA capacitance at common-mode. The disclosed filter arrangement as HT circuit **628** allows over 45 dB of noise rejection at the 4th LO harmonic spur, as shown in FIG. **10**.

(68) In some aspects, when designing the LO 4th harmonic rejection filter of the HT circuit **628**, filter behavior at the LO designed frequency vs. the LO 4th harmonic frequency can be considered. Considerations for both stability and common-mode rejection are a performance tradeoff. Example L-C filter design can start with the assumption that stability has a critical priority. The stability is the main issue that existing solutions cannot implement a high decoupling capacitance at the center tap node, since it will resonate with the transformer inductance at a much lower frequency than LO, creating potential instability. To avoid a significant size decoupling capacitor for noise reduction, the HT circuit **628** can further include an inductor between the decoupling capacitor and the bias circuit to ensure PA stability while significantly reducing LO 4th harmonic common-mode noise. In some aspects, the feedback impedance at ω04 (or the frequency at the LO 4th harmonic) is designed to meet 0Ω, insuring all the CM feedback noise is shorted at this impedance to ground.

(69) In some embodiments, an inductor value of approximately 400 pH for L.sub.HF can be used at the secondary inductor **622** center-tap of the inter-stage matching. In some aspects, the inductance value is determined by simulations and determining an optimal tradeoff between PA stability and harmonic noise rejection. In some aspects, a decoupling capacitor C.sub.HF is added at the center-tap of inter-stage matching as well, with a capacitance value also determined by the above analysis. In some aspects, the decoupling capacitor re-tunes at the DC bias provided by the DC bias voltage source **626**.

(70) FIG. **7** illustrates a block diagram **700** of PA input impedance at common mode, in accordance with some embodiments. Referring to FIG. **7**, PA **701** receives DC bias voltage **704** via a center tap of the secondary inductor **702**. Diagrams **706** and **708** illustrate the input impedance of PA **701**, where Ls is the inductance of the secondary inductor **702**, C.sub.GD is the gate-drain capacitance, C.sub.GS is the gate-source capacitance, and Cneut is the neutralization capacitance of PA **701**.
(71) FIG. **8** illustrates a block diagram of RF bias implementations in connection with the RF circuitry of FIG. **5**, in accordance with some embodiments. Referring to FIG. **8**, diagrams **800** and **804** illustrate that the PA receives DC bias voltage, with resulting LO 4.sup.th order harmonic noise **802** being mitigated by the bias capacitor Ca.
(72) Diagrams **806** and **808** illustrate that the PA receives DC bias voltage, with resulting LO 4.sup.th order harmonic noise **807** being mitigated by the L-C filter **810** formed by L.sub.HF and C.sub.HF.
(73) In some embodiments, the following equations can be used for configuring the inductance and capacitance of L.sub.HF and C.sub.HF.
(74) $L_{HF}C_{HF} = \frac{1}{{}_{o4}^{2}}$ and (1) $Z_{\mathrm{fb}} = \frac{1}{4}\ {}_{\mathrm{stab}}L_{S} - \frac{{}_{\mathrm{stab}}L_{HF}}{{}_{\mathrm{stab}}^{2}L_{HF}C_{HF} - 1} = 0,$ (2)
(75) where ω.sub.04 is the frequency of the LO 4.sup.th order harmonic, ω.sub.stab is the stability frequency (e.g., the frequency for the weakest stability, which can be the smallest of the positive real part of the common-mode feedback impedance). The real part of the feedback impedance (which is also illustrated in FIG. **13**) can be configured as a positive value to keep the impedance stable.
(76) From equations (1) and (2), the following equations for the inductance and capacitance of the L-C filter **810** can be derived:
(77) $L_{HF} = \frac{L_{s}}{4\left(\frac{{}_{\mathrm{stab}}^{2}}{{}_{o4}^{2}} - 1\right)}$ and (3) $C_{HF} = \frac{1}{{}_{o4}^{2}L_{HF}}.$ (4)
(78) For example, the inductance of the secondary inductor **622** of transformer **618** can be configured as L.sub.S=2.0 nH, the 4th order harmonic is at 42 GHz, and the stabilized frequency can be designed to be 63 GHz. From equation (3), L.sub.HF=424 pH. From equation (4), the C.sub.HF has the value of 33.9 fF.
(79) FIG. **9** illustrates block diagrams **900** and **908** of RF bias implementations in connection with the RF circuitry of FIG. **5**, including multi-pole and adjustable pole harmonic filters, in accordance with some embodiments. Referring to FIG. **9**, diagram **900** illustrates filter arrangement **902** configured as multi-pole LO 4.sup.th order harmonic filter comprising multiple L-C filters **904**, **906**, etc. Diagram **908** illustrates filter arrangement **910** configured as an adjustable pole LO 4.sup.th order harmonic filter comprising a single inductor **912** with multiple, switchable unit capacitors **914**.
(80) FIG. **10** is a graph **1000** of common-mode trans-impedance showing the effect of a harmonic rejection filter, in accordance with some embodiments. As illustrated in FIG. **10**, HT circuit **105** can be configured to provide harmonic rejection **1002** at a preconfigured frequency (e.g., 42 GHz) with over 40 dB attenuation.
(81) FIG. **11** is graph **1100** of the feedback impedance Z.sub.FB showing the effect of a harmonic rejection filter, in accordance with some embodiments.
(82) FIG. **12** is a diagram **1200** illustrating common-mode feedback impedance at the PA **1202** of RF circuitry (e.g., the RF circuitry of FIG. **6**), in accordance with some embodiments. More specifically, FIG. **12** shows the common-mode feedback impedance at node D of PA **1202**, which is at the drain terminals of transistor pair **1206** (e.g., transistors NP and NN). Filter **1204** is configured as a LO 4th order harmonic blocking tank, formed by C.sub.HF and L.sub.HF. Inductor **1212** is the equivalent inductor formed by the inductance Ls of the secondary inductor **622** of transformer **618**.
(83) Z.sub.FB is the common-mode impedance looking back into the balun output node of PA **1202**. Capacitance Cgs is the parasitic capacitance of the gate and source terminals of NP or NN,

and Cgd is the parasitic capacitance of the gate and drain terminals of NP or NN. Cneut is the neutralization capacitance of the power amplifier transconductance stage. Capacitor **1216** C.sub.LF is the DC bias low-frequency noise filtering capacitor.
(84) Cgd′ (combined Cgd and Cneut, Cgd′=Cgd+Cneu) forms feedback from the drain terminals of NP and NN to the gate terminals of NP and NN and may cause instability. To investigate the stability, the impedance **1208** looking into the drain of NP and NN (Zd) is derived in Equation (5) below. The oscillation occurs when the real part of Zd is negative and the imaginary part of Zd crosses zero. Matlab simulation can be used to illustrate that R.sub.HF **1214** can be added to improve stability.
(85) FIG. **13** is a graph **1300** of simulated common-mode feedback impedance (Zd) associated with the RF circuitry in FIG. **12**, in accordance with some embodiments.
(86) If R.sub.HF=0, as shown in FIG. **13**, the feedback impedance is unstable around 80 GHz. If R.sub.HF=3.5 Ohm, which is parasitic resistance of L.sub.HF combined with an in-series designed resistor (total is R.sub.HF), it becomes stable as shown in FIG. **13**.
(87) The following equations may be used for configuring R.sub.HF:
(88) $Z_d = 0.5\frac{0.5 + sC'_{gd} Z_b}{sC'_{gd}(0.5 + Z_b g_m)} = \frac{0.5}{sC'_{gd}}$ .Math. $\frac{0.5G_b + sC'_{gd}}{0.5G_b + g_m}$, (5)
(89) where Z.sub.b **1210** is the Z.sub.FB parallel with the capacitor C.sub.gs.
(90) Additionally,
(91) $Z_b = \frac{Z_{FB}}{1 + 2sC_{gs} Z_{FB}}$ and (6) $G_b = \frac{1}{Z_b}$,
where s is the Laplace variable and g.sub.m is the transconductance of transistor pair **1206**.
(92) In this regard, Equation (5) can be used to perform the Zd simulation by Matlab, and then select a small value for R.sub.HF to let the real part of Zd be positive (as shown in FIG. **13**). The positive real part of Zd keeps the circuit stable. The reason to select a small value of R.sub.HF (e.g., below a threshold established by simulation) is to keep the quality factor of the filter tank (e.g., HT circuit **628**) formed by L.sub.HF and C.sub.HF optimal so it reduces the common-mode LO 4th harmonic noise.
(93) FIG. **14** illustrates a flow diagram of a method **1400** for harmonic trapping in a matching network of a power amplifier, in accordance with some embodiments. Referring to FIG. **14**, method **1400** includes operations **1402**, **1404**, **1406**, and **1408**, which may be executed by control circuitry of a wireless device (e.g., hardware processor **1502** of machine **1500** illustrated in FIG. **15**).
(94) At operation **1402**, the primary inductance and secondary inductance of a differential transformer of the matching network are determined, based on a signal operating frequency of the power amplifier. For example, primary inductance Lp and secondary inductors Ls are determined for corresponding primary inductor **620** and secondary inductor **622** of transformer **618**.
(95) At operation **1404**, an inductance value for an L-C filter is determined, based on the secondary inductance and a harmonic frequency of a local oscillator (LO) signal. For example and about Equations (1)-(4) above, L.sub.HF can be determined based on Ls, the stabilized frequency, and the LO 4.sup.th order harmonic frequency.
(96) At operation **1406**, a capacitance value for the L-C filter is determined, based on the inductance value and the harmonic frequency of the LO signal. For example and about Equations (1)-(4) above, C.sub.HF can be determined based on L.sub.HF and the LO 4.sup.th order harmonic frequency.
(97) At operation **1408**, the L-C filter is provided on an electric connection between a direct current (DC) bias voltage source and a secondary inductor of the differential transformer. For example, the L-C filter of HT circuit **628** is configured on the electric connection between the DC bias voltage source **626** and the secondary inductor **622** of transformer **618**. The L-C filter can be configured with the determined inductance value and the determined capacitance value.

(98) FIG. **15** illustrates a block diagram of an example machine **1500** upon which any one or more of the techniques (e.g., methodologies) discussed herein may perform. In alternative embodiments, the machine **1500** may operate as a standalone device or may be connected (e.g., networked) to other machines. In a networked deployment, machine **1500** may operate in the capacity of a server machine, a client machine, or both in server-client network environments. In an example, the machine **1500** may act as a peer machine in a peer-to-peer (P2P) (or other distributed) network environment. The machine **1500** may be a personal computer (PC), a tablet PC, a set-top box (STB), a personal digital assistant (PDA), a portable communications device, a mobile telephone, a smartphone, a web appliance, a network router, switch or bridge, or any machine capable of executing instructions (sequential or otherwise) that specify actions to be taken by that machine. Further, while only a single machine is illustrated, the term “machine” shall also be taken to include any collection of machines that individually or jointly execute a set (or multiple sets) of instructions to perform any one or more of the methodologies discussed herein, such as cloud computing, software as a service (SaaS), other computer cluster configurations.
(99) Machine (e.g., computer system) **1500** may include a hardware processor **1502** (e.g., a central processing unit (CPU), a graphics processing unit (GPU), a hardware processor core, or any combination thereof), a main memory **1504**, and a static memory **1506**, some or all of which may communicate with each other via an interlink (e.g., bus) **1508**.
(100) Specific examples of main memory **1504** include Random Access Memory (RAM), and semiconductor memory devices, which may include, in some embodiments, storage locations in semiconductors such as registers. Specific examples of static memory **1506** include non-volatile memory, such as semiconductor memory devices (e.g., Electrically Programmable Read-Only Memory (EPROM), Electrically Erasable Programmable Read-Only Memory (EEPROM)) and flash memory devices; magnetic disks, such as internal hard disks and removable disks; magneto-optical disks; RAM, and CD-ROM and DVD-ROM disks.
(101) Machine **1500** may further include a display device **1510**, an input device **1512** (e.g., a keyboard), and a user interface (UI) navigation device **1514** (e.g., a mouse). In an example, the display device **1510**, input device **1512**, and UI navigation device **1514** may be a touch screen display. The machine **1500** may additionally include a storage device (e.g., drive unit or another mass storage device) **1516**, a signal generation device **1518** (e.g., a speaker), a network interface device **1520**, and one or more sensors **1521**, such as a global positioning system (GPS) sensor, compass, accelerometer, or other sensors. The machine **1500** may include an output controller **1528**, such as a serial (e.g., universal serial bus (USB), parallel, or other wired or wireless (e.g., infrared (IR), near field communication (NFC), etc.) connection to communicate or control one or more peripheral devices (e.g., a printer, card reader, etc.). In some embodiments, the processor **1502** and/or instructions **1524** may comprise processing circuitry and/or transceiver circuitry.
(102) The storage device **1516** may include a machine-readable medium **1522** on which is stored one or more sets of data structures or instructions **1524** (e.g., software) embodying or utilized by any one or more of the techniques or functions described herein. The instructions **1524** may also reside, completely or at least partially, within the main memory **1504**, within static memory **1506**, or within the hardware processor **1502** during execution thereof by the machine **1500**. In an example, one or any combination of the hardware processor **1502**, the main memory **1504**, the static memory **1506**, or the storage device **1516** may constitute machine-readable media.
(103) Specific examples of machine-readable media may include non-volatile memory, such as semiconductor memory devices (e.g., EPROM or EEPROM) and flash memory devices; magnetic disks, such as internal hard disks and removable disks; magneto-optical disks; RAM; and CD-ROM and DVD-ROM disks.
(104) While the machine-readable medium **1522** is illustrated as a single medium, the term “machine-readable medium” may include a single medium or multiple media (e.g., a centralized or distributed database, and/or associated caches and servers) configured to store one or more

instructions **1524**.

(105) An apparatus of the machine **1500** may be one or more of a hardware processor **1502** (e.g., a central processing unit (CPU), a graphics processing unit (GPU), a hardware processor core, or any combination thereof), a main memory **1504** and a static memory **1506**, one or more sensors **1521**, a network interface device **1520**, antennas **1560**, a display device **1510**, an input device **1512**, a UI navigation device **1514**, a storage device **1516**, instructions **1524**, a signal generation device **1518**, and an output controller **1528**. The apparatus may be configured to perform one or more of the methods and/or operations disclosed herein. The apparatus may be intended as a component of the machine **1500** to perform one or more of the methods and/or operations disclosed herein, and/or to perform a portion of one or more of the methods and/or operations disclosed herein. In some embodiments, the apparatus may include a pin or other means to receive power. In some embodiments, the apparatus may include power conditioning hardware.

(106) The term "machine-readable medium" may include any medium that is capable of storing, encoding, or carrying instructions for execution by the machine **1500** and that cause the machine **1500** to perform any one or more of the techniques of the present disclosure, or that is capable of storing, encoding or carrying data structures used by or associated with such instructions. Non-limiting machine-readable medium examples may include solid-state memories and optical and magnetic media. Specific examples of machine-readable media may include non-volatile memory, such as semiconductor memory devices (e.g., Electrically Programmable Read-Only Memory (EPROM), Electrically Erasable Programmable Read-Only Memory (EEPROM)) and flash memory devices; magnetic disks, such as internal hard disks and removable disks; magneto-optical disks; Random Access Memory (RAM); and CD-ROM and DVD-ROM disks. In some examples, machine-readable media may include non-transitory machine-readable media. In some examples, machine-readable media may include machine-readable media that is not a transitory propagating signal.

(107) The instructions **1524** may further be transmitted or received over a communications network **1526** using a transmission medium via the network interface device **1520** utilizing any one of a number of transfer protocols (e.g., frame relay, internet protocol (IP), transmission control protocol (TCP), user datagram protocol (UDP), hypertext transfer protocol (HTTP), etc.). Example communication networks may include a local area network (LAN), a wide area network (WAN), a packet data network (e.g., the Internet), mobile telephone networks (e.g., cellular networks), Plain Old Telephone (POTS) networks, and wireless data networks (e.g., Institute of Electrical and Electronics Engineers (IEEE) 802.11 family of standards known as Wi-Fi®, IEEE 802.16 family of standards known as WiMax®), IEEE 802.15.4 family of standards, a Long Term Evolution (LTE) family of standards, a Universal Mobile Telecommunications System (UMTS) family of standards, peer-to-peer (P2P) networks, among others.

(108) In an example, the network interface device **1520** may include one or more physical jacks (e.g., Ethernet, coaxial, or phone jacks) or one or more antennas to connect to the communications network **1526**. In an example, the network interface device **1520** may include one or more antennas **1560** to wirelessly communicate using at least one single-input multiple-output (SIMO), multiple-input multiple-output (MIMO), or multiple-input single-output (MISO) techniques. In some examples, the network interface device **1520** may wirelessly communicate using Multiple User MIMO techniques. The term "transmission medium" shall be taken to include any intangible medium that is capable of storing, encoding, or carrying instructions for execution by the machine **1500**, and includes digital or analog communications signals or other intangible media to facilitate communication of such software.

(109) Examples, as described herein, may include, or may operate on, logic or a number of components, modules, or mechanisms. Modules are tangible entities (e.g., hardware) capable of performing specified operations and may be configured or arranged in a certain manner. In an example, circuits may be arranged (e.g., internally or concerning external entities such as other

circuits) in a specified manner as a module. In an example, the whole or part of one or more computer systems (e.g., a standalone, client, or server computer system) or one or more hardware processors may be configured by firmware or software (e.g., instructions, an application portion, or an application) as a module that operates to perform specified operations. In an example, the software may reside on a machine-readable medium. In an example, the software, when executed by the underlying hardware of the module, causes the hardware to perform the specified operations.
(110) Accordingly, the term “module” is understood to encompass a tangible entity, be that an entity that is physically constructed, specifically configured (e.g., hardwired), or temporarily (e.g., transitorily) configured (e.g., programmed) to operate in a specified manner or to perform part or all of any operation described herein. Considering examples in which modules are temporarily configured, each of the modules need not be instantiated at any one moment in time. For example, where the modules comprise a general-purpose hardware processor configured using the software, the general-purpose hardware processor may be configured as respective different modules at different times. The software may accordingly configure a hardware processor, for example, to constitute a particular module at one instance of time and to constitute a different module at a different instance of time.
(111) Some embodiments may be implemented fully or partially in software and/or firmware. This software and/or firmware may take the form of instructions contained in or on a non-transitory computer-readable storage medium. Those instructions may then be read and executed by one or more processors to enable the performance of the operations described herein. The instructions may be in any suitable form, such as but not limited to source code, compiled code, interpreted code, executable code, static code, dynamic code, and the like. Such a computer-readable medium may include any tangible non-transitory medium for storing information in a form readable by one or more computers, such as but not limited to read-only memory (ROM); random access memory (RAM); magnetic disk storage media, optical storage media; flash memory, etc.
(112) The above-detailed description includes references to the accompanying drawings, which form a part of the detailed description. The drawings show, by way of illustration, specific embodiments that may be practiced. These embodiments are also referred to herein as “examples.” Such examples may include elements in addition to those shown or described. However, also contemplated are examples that include the elements shown or described. Moreover, also contemplated are examples using any combination or permutation of those elements shown or described (or one or more aspects thereof), either with respect to a particular example (or one or more aspects thereof) or with respect to other examples (or one or more aspects thereof) shown or described herein.
(113) Publications, patents, and patent documents referred to in this document are incorporated by reference herein in their entirety, as though individually incorporated by reference. In the event of inconsistent usages between this document and those documents so incorporated by reference, the usage in the incorporated reference(s) are supplementary to that of this document; for irreconcilable inconsistencies, the usage in this document controls.
(114) In this document, the terms “a” or “an” are used, as is common in patent documents, to include one or more than one, independent of any other instances or usages of “at least one” or “one or more.” In this document, the term “or” is used to refer to a nonexclusive or, such that “A or B” includes “A but not B,” “B but not A,” and “A and B,” unless otherwise indicated. In the appended claims, the terms “including” and “in which” are used as the plain-English equivalents of the respective terms “comprising” and “wherein.” Also, in the following claims, the terms “including” and “comprising” are open-ended, that is, a system, device, article, or process that includes elements in addition to those listed after such a term in a claim are still deemed to fall within the scope of that claim. Moreover, in the following claims, the terms “first,” “second,” and “third,” etc. are used merely as labels and are not intended to suggest a numerical order for their objects.

(115) The embodiments as described above may be implemented in various hardware configurations that may include a processor for executing instructions that perform the techniques described. Such instructions may be contained in a machine-readable medium such as a suitable storage medium or a memory or other processor-executable medium.
(116) The embodiments as described herein may be implemented in a number of environments such as part of a wireless local area network (WLAN), 3rd Generation Partnership Project (3GPP) Universal Terrestrial Radio Access Network (UTRAN), or Long-Term-Evolution (LTE) or a Long-Term-Evolution (LTE) communication system, although the scope of the disclosure is not limited in this respect.
(117) Antennas referred to herein may comprise one or more directional or omnidirectional antennas, including, for example, dipole antennas, monopole antennas, patch antennas, loop antennas, microstrip antennas, or other types of antennas suitable for transmission of RF signals. In some embodiments, instead of two or more antennas, a single antenna with multiple apertures may be used. In these embodiments, each aperture may be considered a separate antenna. In some multiple-input multiple-output (MIMO) embodiments, antennas may be effectively separated to take advantage of spatial diversity and the different channel characteristics that may result between each of antennas and the antennas of a transmitting station. In some MIMO embodiments, antennas may be separated by up to 1/10 of a wavelength or more.
(118) Described implementations of the subject matter can include one or more features, alone or in combination as illustrated below by way of examples.
(119) Example 1 is an apparatus for a communication device, the apparatus comprising: baseband circuitry to generate a differential baseband signal using a local oscillator (LO) signal; matching circuitry, comprising: a transformer to receive the differential baseband signal and generate a transformed signal based on the differential baseband signal; and a direct current (DC) bias voltage source, the DC bias voltage source coupled to the transformer via a filter arrangement; and amplification circuitry to amplify the transformed signal.
(120) In Example 2, the subject matter of Example 1 includes subject matter where the filter arrangement is configured to filter out an even-order harmonic of the LO signal from the transformed signal.
(121) In Example 3, the subject matter of Examples 1-2 includes subject matter where the filter arrangement is configured to filter out a fourth-order harmonic of the LO signal from the transformed signal.
(122) In Example 4, the subject matter of Examples 1-3 includes subject matter where the filter arrangement is an L-C filter.
(123) In Example 5, the subject matter of Example 4 includes subject matter where the L-C filter is configured with capacitance and inductance values resulting in a rejection frequency associated with a fourth-order harmonic of the LO signal.
(124) In Example 6, the subject matter of Examples 1-5 includes subject matter where the transformer comprises a primary inductor and a secondary inductor, and wherein the DC bias voltage source is coupled to the secondary inductor via an electric connection including the filter arrangement.
(125) In Example 7, the subject matter of Example 6 includes subject matter where the electric connection couples the DC bias voltage source and a center tap of the secondary inductor.
(126) In Example 8, the subject matter of Examples 1-7 includes a grounded capacitor, the grounded capacitor coupled between the filter arrangement and the DC voltage source.
(127) In Example 9, the subject matter of Example 8 includes subject matter where the grounded capacitor is configured to filter low-frequency noise signals generated by the DC bias voltage source.
(128) In Example 10, the subject matter of Examples 1-9 includes subject matter where the filter arrangement comprises a single inductor and a plurality of switchable unit capacitors.

(129) Example 11 is a method for harmonic trapping in a matching network of a power amplifier, the method comprising: determining primary inductance and secondary inductance of a differential transformer of the matching network, based on a signal operating frequency of the power amplifier; determining an inductance value for an L-C filter, based on the secondary inductance and a harmonic frequency of a local oscillator (LO) signal; determining a capacitance value for the L-C filter, based on the inductance value and the harmonic frequency of the LO signal; and providing the L-C filter on an electric connection between a direct current (DC) bias voltage source and a secondary inductor of the differential transformer, the L-C filter configured with the determined inductance value and the determined capacitance value.
(130) In Example 12, the subject matter of Example 11 includes, determining the inductance value further based on a stability frequency, the stability frequency corresponding to a minimal positive real part of a common-mode feedback impedance of the power amplifier.
(131) In Example 13, the subject matter of Example 12 includes, coupling the electric connection between the DC bias voltage source and a center tap of the secondary inductor.
(132) In Example 14, the subject matter of Examples 12-13 includes, providing a grounded capacitor coupled between the L-C filter and the DC voltage source.
(133) In Example 15, the subject matter of Example 14 includes, configuring a capacitance of the grounded capacitor, the capacitance of the grounded capacitor resulting in filtering low-frequency noise signals generated by the DC bias voltage source.
(134) Example 16 is an apparatus for a communication device, the apparatus comprising: a power amplifier stage, the power amplifier stage comprising: a transformer, the transformer driven by a differential baseband signal based on a local oscillator (LO) signal; and a direct current (DC) bias voltage source, the DC bias voltage source coupled to the transformer via a filter arrangement; wherein the filter arrangement comprises at least one inductor associated with an inductance value, the inductance value based on a secondary inductance of the transformer and a harmonic frequency of the LO signal.
(135) In Example 17, the subject matter of Example 16 includes subject matter where the filter arrangement comprises at least one L-C filter, and wherein the at least one L-C filter is associated with a capacitance value, the capacitance value, and the inductance value resulting in a rejection frequency associated with a fourth-order harmonic of the LO signal.
(136) In Example 18, the subject matter of Examples 16-17 includes subject matter where the transformer comprises a primary inductor and a secondary inductor, and wherein the DC bias voltage source is coupled to the secondary inductor via an electric connection including the filter arrangement, wherein the electric connection couples the DC bias voltage source and a center tap of the secondary inductor.
(137) In Example 19, the subject matter of Examples 16-18 includes a grounded capacitor, the grounded capacitor coupled between the filter arrangement and the DC voltage source.
(138) In Example 20, the subject matter of Example 19 includes subject matter where the grounded capacitor is configured to filter low-frequency noise signals generated by the DC bias voltage source.
(139) Example 21 is at least one machine-readable medium including instructions that, when executed by processing circuitry, cause the processing circuitry to perform operations to implement any of Examples 1-20.
(140) Example 22 is an apparatus comprising means to implement any of Examples 1-20.
(141) Example 23 is a system to implement any of Examples 1-20.
(142) Example 24 is a method to implement any of Examples 1-20.
(143) The above description is intended to be illustrative, and not restrictive. For example, the above-described examples (or one or more aspects thereof) may be used in combination with others. Other embodiments may be used, such as by one of ordinary skill in the art upon reviewing the above description. The Abstract is to allow the reader to quickly ascertain the nature of the

technical disclosure. It is submitted with the understanding that it will not be used to interpret or limit the scope or meaning of the claims. Also, in the above Detailed Description, various features may be grouped to streamline the disclosure. However, the claims may not set forth every feature disclosed herein as embodiments may feature a subset of said features. Further, embodiments may include fewer features than those disclosed in a particular example. Thus, the following claims are hereby incorporated into the Detailed Description, with a claim standing on its own as a separate embodiment. The scope of the embodiments disclosed herein is to be determined regarding the appended claims, along with the full scope of equivalents to which such claims are entitled.

Claims

1. An apparatus for a communication device, the apparatus comprising: baseband circuitry to generate a differential baseband signal using a local oscillator (LO) signal; matching circuitry, comprising: a transformer to receive the differential baseband signal and generate a transformed signal based on the differential baseband signal; and a direct current (DC) bias voltage source, the DC bias voltage source coupled to the transformer via a filter arrangement, the filter arrangement configured to reduce a signal level of at least one harmonic of the LO signal from the transformed signal; and amplification circuitry to amplify the transformed signal.
2. The apparatus of claim 1, wherein the at least one harmonic is an even-order harmonic of the LO signal.
3. The apparatus of claim 1, wherein the at least one harmonic is a fourth-order harmonic of the LO signal.
4. The apparatus of claim 1, wherein the filter arrangement is an L-C filter.
5. The apparatus of claim 4, wherein the L-C filter is configured with capacitance and inductance values resulting in a rejection frequency associated with a fourth-order harmonic of the LO signal.
6. The apparatus of claim 1, wherein the transformer comprises a primary inductor and a secondary inductor, and wherein the DC bias voltage source is coupled to the secondary inductor via an electric connection including the filter arrangement.
7. The apparatus of claim 6, wherein the electric connection couples the DC bias voltage source and a center tap of the secondary inductor.
8. The apparatus of claim 1, further comprising a grounded capacitor, the grounded capacitor coupled between the filter arrangement and the DC voltage source.
9. The apparatus of claim 8, wherein the grounded capacitor is configured to filter low-frequency noise signals generated by the DC bias voltage source.
10. The apparatus of claim 1, wherein the filter arrangement comprises a single inductor and a plurality of switchable unit capacitors.
11. A method for harmonic trapping in a matching network of a power amplifier, the method comprising: determining primary inductance and secondary inductance of a differential transformer of the matching network, based on a signal operating frequency of the power amplifier; determining an inductance value for an L-C filter, based on the secondary inductance and a harmonic frequency of a local oscillator (LO) signal; determining a capacitance value for the L-C filter, based on the inductance value and the harmonic frequency of the LO signal; and providing the L-C filter on an electric connection between a direct current (DC) bias voltage source and a secondary inductor of the differential transformer, the L-C filter configured with the determined inductance value and the determined capacitance value.
12. The method of claim 11, further comprising: determining the inductance value further based on a stability frequency, the stability frequency corresponding to a minimal positive real part of a common-mode feedback impedance of the power amplifier.
13. The method of claim 12, further comprising: coupling the electric connection between the DC bias voltage source and a center tap of the secondary inductor.

14. The method of claim 12, further comprising: providing a grounded capacitor coupled between the L-C filter and the DC voltage source.
15. The method of claim 14, further comprising: configuring a capacitance of the grounded capacitor, the capacitance of the grounded capacitor resulting in filtering low-frequency noise signals generated by the DC bias voltage source.
16. An apparatus for a communication device, the apparatus comprising: a power amplifier stage, the power amplifier stage comprising: a transformer, the transformer driven by a differential baseband signal based on a local oscillator (LO) signal; and a direct current (DC) bias voltage source, the DC bias voltage source coupled to the transformer via a filter arrangement; wherein the filter arrangement comprises at least one inductor associated with an inductance value, the inductance value based on a secondary inductance of the transformer, and a harmonic frequency of the LO signal.
17. The apparatus of claim 16, wherein the filter arrangement comprises at least one L-C filter, and wherein the at least one L-C filter is associated with a capacitance value, the capacitance value, and the inductance value resulting in a rejection frequency associated with a fourth-order harmonic of the LO signal.
18. The apparatus of claim 16, wherein the transformer comprises a primary inductor and a secondary inductor, and wherein the DC bias voltage source is coupled to the secondary inductor via an electric connection including the filter arrangement, wherein the electric connection couples the DC bias voltage source and a center tap of the secondary inductor.
19. The apparatus of claim 16, further comprising a grounded capacitor, the grounded capacitor coupled between the filter arrangement and the DC voltage source.
20. The apparatus of claim 19, wherein the grounded capacitor is configured to filter low-frequency noise signals generated by the DC bias voltage source.
